

Prof. Desirée Pacheco
e-mail: dpacheco@iese.edu
Phone: 93 253 6554

Assistant: Denise Clerc
e-mail: dclerc@iese.edu
Office: B-400
Tel: 93 602 4169

Sustainable Entrepreneurship and Innovation

INTRODUCTION

The current state of the natural environment, and prevailing social challenges around the globe, call for innovative action that can transform current paradigms. To guide such transformation, the United Nations has set sustainable development goals (SDGs) that target environmental and social areas for urgent intervention. This course examines key concepts of entrepreneurship, using the SDGs as guiding principles. It explores how new ventures, non-profits, and other companies address and advance the SDGs through their innovation. We examine the business models, strategies, mobilization, and challenges of sustainable enterprises. Students leverage this knowledge to identify and evaluate entrepreneurial opportunities that tackle the SDG(s) of their choice.

OBJECTIVES

- Understand a range of business models with the key mission of addressing one or more of the SDGs.
- Understand the SDGs and identify opportunities for involvement.
- Assess specific sustainable entrepreneurship opportunities through a preliminary feasibility analysis.
- Feel energized and empowered to apply entrepreneurship tools towards environmental and social wellbeing.

CONTENT

The course is built upon the UN's SDGs. It is therefore, designed around environmental and social challenges, and related goals. The underlying philosophy is that we do not merely want to understand business tools and models, and then apply them. Rather, we want to first gain fundamental understanding of the challenges faced and then assess or design models that can help to solve those challenges in meaningful ways.

There are 20 sessions scheduled, 2 will be used for meetings with the professor to discuss team projects.

METHODOLOGY

The course will use a variety of methods to deliver learning outcomes. This will include: In-class discussion of Case Studies, Scientific Reports, and Videos, as well as Interactive Lectures, Guest Speakers, and a Team Project.

EVALUATION

Student evaluation will be a mix of in-class participation (40%), one individual reading analysis (10%), and the Opportunity Assessment Project (50%).

In-Class Engagement and Participation

Please come ready to participate to each session. Participation and contribution to class learning will be determined by preparation for class, active participation in class discussions, and your ability to demonstrate that you have prepared the materials assigned to a particular class, both in terms of quality and quantity (with greater weight on the quality and depth of your answers).

I ask that you are fully present in the classroom, which means not only that you speak, but that you are actively listening to others (show attentiveness and respect); and are therefore, *engaged* in the class conversation.

Opportunity Assessment Team Project

The Project will consist of an Opportunity Assessment of a sustainable venture or related initiative addressing the SDG(s) of the team's choice. For example, a team can choose to focus on the Diversity, Equity, and Inclusion goal. This team will ideate and choose a market opportunity that tackles that goal (e.g., an enterprise that employs disadvantaged labor), considering the unaddressed needs, and the market failures that lead to them. The team will begin by ideating ways in which value can be captured to address the chosen goal. It will then assess the general feasibility of that market opportunity. This will consist of examining the potential market, the industry and competition, together with a preliminary business model design. The final deliverable is an Opportunity Assessment that summarizes the findings from the research and a summary of identified opportunities and risks. To be delivered on a PPT Deck format and a Final Presentation. For more details on the course project, please refer to the Opportunity Assessment Project Description in Virtual Campus.

*The project must consist of a new concept for the team. It cannot be based on prior research or projects from other courses.

Individual Reading Analysis

Students will select one available session topic for which they will write a reflection and analysis of the readings for that day (a sign-up form will be shared during the first week). The analysis will consist of an up to 2-page answer to one of the assignment questions posted for the selected topic (under Discussion Questions in the course outline below). Please provide an informed opinion to the questions chosen, substantiated by valid sources. Think of this analysis as a well-crafted and informed Op-ed piece.

*The analysis is due at 11:59pm on the day before your chosen topic is discussed, via Virtual Campus. Late assignments are not accepted unless arrangements have been made prior to the due date.

There is no required format style for your response (except maximum length), but please keep in mind that quality goes over quantity. Please show that you understand the topic and its' implications for entrepreneurship and the role of markets in addressing your chosen SDG.

LEARNING OUTCOMES

- Understand key environmental and social challenges, and related market and non-market solutions
- Articulate and discuss the circumstances that lead to success or failure in sustainable enterprises.
- Learn how to design and apply different tools for environmental and community impact
- Assess social and environmentally responsible opportunities through a feasibility analysis

COMPETENCES

- Develop a social and environmentally conscious perspective regarding business opportunities
- Develop critical thinking skills around the nature of social and environmental challenges
- Solve environmental and social problems with the use of business tools
- Ability to assess social and environmentally responsible opportunities
- Ideate and innovate with respect to social and environmental challenges

COURSE OUTLINE

Below please find the outline by session for the course, which is subject to change. Any change will be announced in class.

Session 1 September 9	Topic: SDGs, Entrepreneurship, and Business Models for Sustainable Innovation	Discussion Questions: *Please read the course <u>project</u> description before coming to class and come with any questions you may have. -What is sustainable entrepreneurship? How can it be used as a tool to address the SDGs? What SDGs do you think entrepreneurship may be best able to serve? Why? -Please refresh your knowledge of business models and the business model canvas. -How are the business models of sustainable enterprises different from traditional business models?
	<u>Please read and watch:</u> • Business Model Review: Osterwalder and Pigneur, Business Model Generation, Chapter 1. (In Virtual Campus) • Familiarize yourself with the Sustainable Development Goals: https://sdgs.un.org/goals • Video: The Global Goals we've made progress on-the ones we haven't Michael Green: https://www.youtube.com/watch?v=N3SQIrmV1cE	

	<u>Want to Know More?</u> - Christensen, Baumann, Ruggles, and Sadtler, 2006. Disruptive Innovation for Social Change. <i>Harvard Business Review</i>. - The Impact Business Model Canvas (good examples here on how to build a canvas for sustainable enterprises): https://stanford.edu/dept/gsb-ds/Inkling/The_Impact_BMC/index.html - Holburn and Raiha, Startups are Turning Customers into Lobbyists, <i>Harvard Business Review</i>, Oct. 24, 2017.	
Session 2 September 9	Topic: Greyston Bakery Case SDG 10: Reduced Inequality	Deliverable/Discussion Questions: -Why should business be concerned with the effects of economic inequality? What role should entrepreneurship play in addressing these effects? -Can Greyston Bakery's Open Hiring policy be adopted by more traditional companies? What obstacles do you anticipate that companies will face? How can they be addressed? -How would the Business Model Canvas of Greyston Bakery look like? -What industry or company conditions may be best suitable to an Open Hiring policy? Why? -As a Mid-Hudson Electronics manager, how would you present an Open Hiring policy to the board of directors? What aspects would you highlight and how? -What challenges do you think entrepreneurs like Paolo Fusaro of Robin Good face when creating more inclusive ventures? How would you address them to create and capture social value through your venture?
	<u>Readings:</u> <u>Case:</u> Greyston Bakery, available at: https://www.stern.nyu.edu/sites/default/files/assets/documents/Greyston_Van%20Wert_08.2018.pdf - Rosenberg, 2019. No Background Check, Drug Test or Credit Check. You're Hired!, <i>The New York Times</i>, May 29, 2019 <u>Want to know more?</u> - Homeboy Industries: Flanigan, 2008. A Budding Conglomerate is Really a Home for Ex-Gang Members. <i>The New York Times</i>, March 20, 2008. - De Neve and Nattavudh, Income Inequality Makes Whole Countries Less Happy. <i>Harvard Business Review</i>, Jan 12, 2016	
Session 3 September 16	Design Thinking for Sustainable Innovation	*Please come ready with a social and/or environmental problem that your team would like to tackle in the course project. -How can design thinking be applied in sustainable entrepreneurship? -What are the advantages of this approach? What disadvantages do you see?
	<u>Readings:</u> Brown, T., Wyatt, J. 2010. Design Thinking for Social Innovation. <i>Stanford Social Innovation Review</i>. <u>Want to Know More?</u> IDEO U Design Thinking Resources: https://www.ideo.com/pages/design-thinking-resources?_pos=14&_sid=f88bc61df&_ss=r	
Session 4 September 16	Design Thinking for Sustainable Innovation II	Interactive Team Exercise in Class

Session 5 September 23	Climate Change Action SDG 13 Climate Action	Deliverable/Discussion Questions: -If you had \$100, which technologies or solutions addressing climate change would you invest in? Why? Assume you can invest in multiple solutions, how much would you allocate to each? -What is the role of impact investing in addressing climate change? Should philanthropy be used in these investments? What advantages or disadvantages do you see in this approach? -What will it take to grow innovative companies like Global Thermostat? What technical challenges would it face? Will GT be able to scale their processes to provide low-cost solutions needed to grow? -What role should entrepreneurship have on climate adaptation (as opposed to mitigation)? What opportunities do you see?
	<u>Required Video:</u> Climate Action - Paul Hawken on Project Drawdown https://www.youtube.com/watch?v=u7K5M-jaHk8 <u>Readings:</u> Background and Science: - Henderson et al., 2020. Climate Change in 2020: Implications for Business, Harvard Business Publishing (SKIM) Entrepreneurship and Potential Solutions: - Familiarize yourself with the climate change solutions proposed in: https://drawdown.org/solutions <u>Case:</u> Global Thermostat: Closing the Carbon Cycle (Thunderbird). Use the case to familiarize yourself with air capture technology and understand the opportunities and challenges in this space. You may skip the sections on Founders and Chairman and Executive Teams. - Scott et al. 2018. The Investment Gap that Threatens the Planet, Stanford Social Innovation Review (SKIM) <u>Want to Know More?:</u> How much innovation is necessary to see off fossil fuels? The Economist, Jan 25th 2023 Sun Machines, The Economist Generative AI has a clean-energy problem, The Economist, Apr 11th 2024 - Friedman, Want to Save the Earth? We Need a Lot More Elon Musks. The New York Times, Nov. 18th, 2021. - Climate Tech Companies: https://www.greenbiz.com/collections/climate-tech - IPCC Global Warming of 1.5C Report (see executive summaries per chapter), https://www.ipcc.ch/sr15/ - Videos: Unburnable Carbon https://www.youtube.com/watch?v=tFlqZtXMqN8 - Rachael Pike TED Talk https://ed.ted.com/lessons/rachel-pike-the-science-behind-a-climate-headline - Book: Drawdown: The Most Comprehensive Plan Ever Proposed to Reverse Global Warming. Paul Hawken, 2017.	
Session 6 September 23	Climate Change Panel SDG 13 Climate Action	<i>Guest Panel on Climate Change: TBD</i>
Session 7 September 30	Clean Water and Sanitation SDG 1 No Poverty SDG 6 Clean Water and Sanitation	Deliverable/Discussion Questions: -What are the most impactful business models to address poverty? Why? How does it compare to alternative models? -Why do companies fail to address Bottom of the Pyramid (BOP) markets? What can be learned from that? Bring an example of a company that faced challenges in addressing a BOP market and come

		ready to explain how you think the company could overcome those obstacles. -What can be learned from Sanergy's early success in addressing BOP markets? -What key factors should Sanergy consider in scaling its' business? -What should companies like Sanergy consider when developing their own value chains in BOP markets?
	<u>Readings:</u> <u>Case:</u> Sanergy: Tackling Sanitation in Kenyan Slums, Berkeley Haas Case Series Kasturi and Gregg, The Scope of Business at the Base of the Pyramid: Middle and Low Income Countries, Harvard Business Publishing, Jan. 16, 2018 (pages 1-12). <u>Want to Know More?</u> Simanis and Duke, Profits at the Bottom of the Pyramid. Harvard Business Review, October 2014. Chakravorti, B. Finding Competitive Advantage in Adversity, Harvard Business Review, Nov. 2010 Bloom and Chatterji, Scaling Social Entrepreneurial Impact, California Management Review, 51(3): 114-133.	
Session 8 September 30	No Poverty SDG 1 No Poverty SDG 8 Decent Work and Economic Growth	Deliverable/Discussion Questions: <u>Case A:</u> -What challenges did the Tony's team face when building the company? How were they solved? How do these compare with more conventional or traditional startups? -What were Tony's strategic priorities in setting up the company? Do you agree with these? Why or why not? -What should Tony's do next? How can the company enhance its' social impact? <u>Case B:</u> -What should the go-to market strategy of Tony's Chocolonely be for the US market, considering the company's strategy and current financial position? -Do you believe that entering the US market was the appropriate growth strategy for Tony's? Why? If not, what alternative growth strategy would you recommend? -How does Tony's growth strategy compare to that of more conventional companies? Do you know of similar companies who have chosen to grow this way?
	<u>Case:</u> Tony's Chocolonely (A) and PART (B) <u>Want to Know More?</u> https://www.youtube.com/watch?v=9Lwvpu57IDU	
Session 9 October 7	Crafting a Sustainability Strategy SDG 15 Life on Land SDG 13 Climate Action SDG 8 Decent Work and Economic Growth	Deliverable/Discussion Questions: -How should Vallformosa outline and structure a cohesive sustainability strategy? What priorities should they set? What key metrics should they track based on these priorities? -How can Vallformosa ensure the wellbeing of winegrowers in the community and the continuation of the profession? -How should Vallformosa continue to address the challenges posed by climate change in the region? -Is the B-Impact Assessment a complete guide to measuring Vallformosa's impact? Are there other metrics or methods that the

		company should use to evaluate progress on their sustainability strategy?
	<u>Readings:</u> Case: Bodegues Vallformosa -B Impact Assessment for Vallformosa (SKIM)	
Session 10 October 7	Impact Strategy and Measurement	Preparation Questions: -What methodologies are most utilized to measure ESG performance? What are their advantages and disadvantages? -How do these tools differ from the measurement of "impact"? -What considerations should entrepreneurs and small enterprises take when measuring impact? What should their initial approach be?
		<u>Readings:</u> Andreou, N.& Besharov, M. Rethinking How We Measure Companies on Social and Environmental Impact, Sloan Management Review, Winter 2023. <u>Want to Know More?</u> Challenges and Opportunities in Social Impact Measurement, Management Solutions, 2022. Available at: https://www.managementsolutions.com/en/microsites/whitepapers/social-impact
Session 11&12 October 14	Project Update Team Meetings with Professor. Schedule to be announced. Please come prepared to share progress on your team projects by this date, including your latest Business Model Canvas. What are the most important insights gained up to date? How is this driving the direction of the project? <u>Due:</u> Submit Project BMC in Virtual Campus by 11:59pm on November 14th (1 per team)	
Session 13 November 11	Impact Investing	<u>Guest Speaker:</u> Xabier Pont
	The Ultimate Introductory Guide to Funding Your Social Enterprise (SKIM): https://blog.acumenacademy.org/impact-capital-funding-your-social-enterprise <u>Want to Know More?</u> Heitmann, K., Lonneke, R., Boiardi, P., & Serneels S. The Rise of the Corporate Social Investor, Stanford Social Innovation Review, Summer 2020.	
Session 14 November 11	Topic: Sustainable Food Production SDG 15 Life on Land SDG 13 Climate Action	Deliverable/Discussion Questions: -How might important stakeholders like customers, investors, retailers, and competitors react to Tyson's entry into the non-animal protein? Why do you anticipate such a reaction? What are the implications for Tyson's strategy? -How does each investment option impact Tyson's branding and marketing strategy? How can the company avoid cannibalizing itself? -What are the environmental and/or social implications of each of the options that Tyson is considering? -Would you invest in the alternatives proteins industry? Why or why not? If yes, where would you put your money? What is your rationale for this? -How do you foresee the future of agriculture? Where do you see the greatest potential to invest for achieving sustainable outcomes? Why? -What role does corporate entrepreneurship in the form of Corporate Venture Capital play in delivering sustainable innovations?
	<u>Readings/Videos:</u> • <u>Case:</u> Tyson Foods and Alternative Proteins: Where to invest for sustainable growth? W97C56	

	- General Information on Alternative Proteins: https://www.cleantech.com/alternative-proteins-and-new-business-models-the-changing-face-of-food/ (SKIM) - Watch: Market Forces and Food Technology Will Save the World Bruce Friedrich https://www.youtube.com/watch?v=liZtyP2tKhA Want to Know More? - Why Investors Need To Take The Long View On Alternative Proteins. Forbes, 2023. https://www.forbes.com/sites/forbesfinancecouncil/2023/06/23/why-investors-need-to-take-the-long-view-on-alternative-proteins/?sh=3f3bf96c4db1 - Startups are Producing Real Dairy without a Cow in Sight. The Economist. https://www.economist.com/business/2023/07/20/startups-are-producing-real-dairy-without-a-cow-in-sight - Intro to Regenerative Agriculture: https://www.nrdc.org/stories/regenerative-agriculture-101#what-is	
Session 15 November 18	Consumption and Waste SDG 9 Industry, innovation and infrastructure SDG 12 Responsible consumption and production	Deliverable/Discussion Questions: -What are some of the most impactful ways to address circularity? Which practices, business models, and companies do you believe are at the forefront of the circular economy? -How should companies position themselves strategically to address and market circularity? How can customers and others in the value chain (e.g., suppliers) get on board? -What is the role of consumer brands like H&M in developing more sustainable supply chains? Who should bear the costs? How should smaller players like Dakota address this? -What are the positive and negative implications of Dakota investing in Texloop without further deliberation? What should it seek from H&M? Is it prepared to handle losses? -Examine the trend of sustainable apparel. What role does entrepreneurship play in it? Are consumers willing to pay more for sustainable fashion because it makes them feel good? Is there evidence of this?
Session 15 November 18	<u>Readings:</u> <u>Case:</u> Dakota Industrial Co. Ltd: Sustainable Garment Manufacturing in a Fast Fashion World, HBP: ST87 - Esposito et al., Introducing a circular economy: new thinking with new managerial and policy implications, California Management Review, 60(3): 5-19. (FOCUS on business model section) - Dean and Pacheco, Green marketing: A strategic balancing act for creating value, Journal of Business Strategy, Sept. 2014. (SKIM) <u>Want to Know More?</u> - What goes around. <i>The Economist</i>, Vol. 428, Iss. 9111, (Sep 29, 2018): S11-S12. - Atasu et al. Rethinking sustainability in light of the EU's new circular economy policy. Harvard Business Review, July 3, 2018 Many resources on the Circular Economy, including industry-specific info at: https://ellenmacarthurfoundation.org/	
Session 16 November 18	<u>TBA</u>	
Session 17 November 25	Topic: Gender Equality SDG 5 Gender Equality	Deliverable/Discussion Questions: -What are the assumptions behind Harlem Capital's Central Investment Thesis? Are these assumptions valid? -What is the market opportunity for Harlem Capital like? What are the major risks?

		-If you were in Jarriid's and Henri's place, would you put your future career at risk and start Harlem Capital? Why or why not?
	<u>Readings:</u> <u>Case:</u> Harlem Capital: Changing the Face of Entrepreneurship (A), HBSP, 9-120-040 -Trends in Female-Founder Venture Funding: https://www.ubs.com/global/en/wealth-management/women/insights/2021/funding-gap.html <u>Want to know more?</u> Haridasani, 2021. Funding for startups founded by women is surging. The New York Times, November 2, 2021	
Session 18 November 25	Topic: Gender Equality and Course Wrap Up	<u>Guest Speaker: Olga Santos</u>
Sessions 19 & 20 December 2	Final Project Presentations	Come ready to present your opportunity assessment project. Details will be shared separately. Two investors from the Barcelona community will join the presentations to provide feedback. <u>Guest Investors:</u> -Cristina Marsal, Partner & CEO, Sandman Capital, (www.sandmancapitalpartners.com) <u>Due:</u> Submit Final Team Pitch Deck in Virtual Campus by 12:00pm on December 9th (1 per team)

Additional Resources:

Resources/Examples of Organizations and Innovations in Sustainable Entrepreneurship:

Cleantech Group. 2018. *Global Cleantech 100: A Barometer of the Changing Face of Global Cleantech Innovation*.

Skim, and review companies in Appendix 3 (p 44-48).

http://info.cleantech.com/GlobalCleantech100Report2018_GCT100Report2018Download.html?aiid=75233532

www.greenbiz.com/enewsletter/ - GreenBuzz Newsletter

www.cleantech.com – Cleantech Venture Group

www.cleantech.com - Clean Edge, Inc.

www.e2.org - Environmental Entrepreneurs

<https://www.theblueeconomy.org/innovations.html>

Technology Solutions by type of problem : <https://solarimpulse.com/solutions-explorer>

Climate Tech Companies: <https://www.greenbiz.com/collections/climate-tech>

Resources in Funding Sustainable Ventures:

Ship2b Investment:

<https://www.ship2b.org/en/investment/>

Fifteen Social Venture Capital firms You Should Know about

<http://causecapitalism.com/15-social-venture-capital-firms-that-you-should-know-about/>

Green VC: List of Green and Social Venture Funding Sources.

<http://greenvc.org/green-and-social-venture-capital/>

Skoll Foundation Awards: <https://skoll.org/about/skoll-awards/>

<https://www.cnbc.com/impact-investing/>

<https://thegiin.org/news-and-events/>

<https://www.ft.com/impact-investing>